

Lesson 22 — Counting Up

Matches LEGO Teacher Portal lesson: "Counting Creatures".

Objective (I can...): I can change a variable's value to count events. **CS concept:** Updating a variable (increment) · **Builds on:** Lesson 21 (creating/storing a variable).

AI Connection (2 min): AI keeps a running tally — how many examples it has seen, how many it got right. *First-principles question:* "What would a robot count to know it's improving?"

Materials: LEGO CS & AI kit + device per pair, trigger object/button. **Model reuse:** Variables model from Lesson 21 — code only.

60-minute flow - Warm-up (8): Tally clap — coach adds a tally per clap; connect to "add 1 to the variable." - **Concept + demo (10):** Demo "change score by 1" inside an event so each press adds one; show the running total. - **Build/Code (30):** Pairs program a counter: each event (press or sensor) adds 1 to a variable and displays it. *Swap at 15 min.* - **Challenge (7):** Add a reset event that sets the counter back to 0. - **Share + pack-up (5):** Pairs demo counting up and resetting. Model stays built.

Pair roles: Driver codes the counter, Navigator triggers events and reads the total. Swap at 15 min. (*Trio rotates code/trigger/read.*)

Differentiation: *Support* — provide the event; students add the "change by 1." *Extension / fast-finisher* — count by 2 or track two counters. *Reteach* — press once, watch it go 0→1, then let them predict the next.

Exit check: Pair's variable increases correctly on each event and can be reset. **At-home (optional):** Count something for a minute (steps, claps); record the number. ~10 min.
